

SWP391 — SOFTWARE PROJECT MANAGEMENT

MedBooking

Online Medical Appointment Booking System

Research Paper

Formatted in Springer LNCS Style

Project: Healthcare Appointment Booking System
Team: Nhóm 1 — SWP391
Version: v1.0
Date: June 2026

Team Members

No.	Full Name	Student ID
1	Nguyen Dac Dung	DE190205
2	Nguyen Van Duc	DE190388
3	Chu Cao Huy	DE190541
4	Nguyen Minh Trung	DE190522
5	Mai Van Luong	DE190512

MedBooking: An Online Medical Appointment Booking System

Integrating AI Chatbot and Telemedicine

Nguyen Dac Dung¹, Nguyen Van Duc¹, Chu Cao Huy¹, Nguyen Minh Trung¹,
and Mai Van Luong¹

FPT University, Danang, Vietnam
{DE190205, DE190388, DE190541, DE190522, DE190512}@fpt.edu.vn
<https://github.com/dung12122005/project-course-team-7>

Abstract. The increasing demand for efficient healthcare services has exposed critical shortcomings in traditional appointment systems: fragmented scheduling across multiple providers, long wait times, and the absence of real-time availability information. This paper presents **MedBooking**, a web-based medical appointment booking platform developed at FPT University as part of the SWP391 capstone project. The system provides patients with centralised scheduling, real-time slot visibility, AI-powered symptom triage, integrated telemedicine, and automated multi-channel reminders with one-click cancellation. Built on a React.js frontend, a Node.js/Express.js backend, and a MongoDB database, MedBooking is the only platform among its Vietnamese counterparts to combine booking, telemedicine, AI chatbot, e-pharmacy, OCR prescription import, and an interactive medical map in a single application. A review of peer-reviewed papers from Springer confirms that all major design decisions are grounded in empirically validated findings on digital health adoption and patient-centred scheduling.

Keywords: Medical Appointment Booking · Telemedicine · AI Chatbot · Healthcare Information System · Digital Health · Patient-Centred Care

1 Introduction

Healthcare accessibility remains a global challenge. In Vietnam, patients frequently navigate fragmented booking channels — calling multiple clinics, waiting for manual confirmations, and receiving no automated follow-up — resulting in high no-show rates and poor user satisfaction. A study on MRI scheduling found that two-thirds of patients waited more than one month for an appointment, with the absence of a centralised booking point as the primary barrier [8].

Digital health platforms have emerged as a promising solution. Yet research consistently shows that perceived usefulness and ease of use are the two strongest predictors of adoption, particularly among older adults [1]. Systems that are technically capable but difficult to use fail in practice [5].

Competitive gap. Table 3 (Section 3) shows that existing Vietnamese platforms each cover at most three of six key feature areas. BookingCare offers booking and a medical map but lacks telemedicine and AI; eDoctor adds telemedicine and e-pharmacy but has no chatbot or map. No single platform combines all six capabilities.

Contribution. MedBooking addresses this gap by delivering:

1. A four-step centralised booking flow with real-time slot availability (FR-10 to FR-13).
2. An AI chatbot for symptom-based specialty recommendation (FR-33).
3. Integrated telemedicine via WebRTC video call and Socket.io chat (FR-30, FR-31).
4. Evidence-based automated reminders with one-click cancel/reschedule links (FR-16).
5. OCR prescription import and e-pharmacy integration (FR-32, FR-34).
6. A Google Maps medical facility locator (FR-35).

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 describes the system architecture and features. Section 4 discusses alignment between the literature and our design. Section 5 concludes with future directions.

2 Related Work

2.1 Telemedicine Adoption and the TAM Model

He et al. [1] surveyed older adults in China using the Technology Acceptance Model (TAM) combined with emotional design theory, analysed via structural equation modelling. *Perceived usefulness* and *perceived ease of use* were the strongest adoption predictors; technological anxiety had a significant negative effect [1]. These results informed MedBooking’s UX priorities: a four-step booking flow, minimum font size, and clear contrast.

Rafi [5] conducted a mixed-methods study with elderly participants. Although a significant portion reported using at least one AI tool, low digital literacy and lack of trust remained major barriers [5]. The study directly motivated MedBooking’s proxy booking feature, which allows users to manage appointments on behalf of less tech-savvy relatives.

2.2 Centralised Scheduling and Wait-Time Reduction

Singer et al. [8] analysed scheduling experiences, noting that long wait times delay diagnosis and treatment. The root cause was often the absence of a unified booking point, leading to a strong recommendation for a centralised contact hub to streamline the process [8]. This constitutes the core architectural principle of MedBooking.

Kaluza et al. [2] modelled online booking systems within a multinomial logit framework. A key finding emerged: exposing *all* available slots can counterintuitively reduce bookings because users infer low demand and lower quality from too many empty slots [2]. Their recommendation to expose only a controlled subset of slots directly shapes MedBooking’s slot-display strategy.

2.3 Evidence-Based Reminder Design

Selim et al. [3] ran an experiment evaluating consumer preferences for health-care appointment reminders. The analysis identified timely delivery before the appointment and a direct cancellation/reschedule link as critical components to increase acceptance and reduce no-shows [3]. MedBooking implements exactly these attributes.

2.4 AI, NLP, and Clinical Decision Support

Sharma [6] reviewed AI, robotics, and NLP applications advancing telemedicine and remote patient monitoring. The study demonstrates that NLP enables efficient symptom triage, personalised recommendations, and virtual consultation support [6]. This constitutes the theoretical foundation for MedBooking’s AI chatbot module.

Silva-Ferreira et al. [7] conducted a scoping review of AI studies in healthcare contexts. A key finding is that ethical, legal, and data-privacy barriers — not just technical limitations — are the primary obstacles to adoption [7]. MedBooking addresses these concerns through encrypted authentication, password hashing, and secure token storage.

2.5 Stakeholder Requirements and System Quality

Ostadmohammadi et al. [4] analysed perceptions of electronic appointment systems across multiple stakeholder groups using qualitative content analysis. Problems fell into both management and non-management categories, showing that overall satisfaction depends heavily on both functional performance and user experiences [4]. MedBooking addresses this by balancing comprehensive features with strict non-functional requirements.

3 System Architecture and Key Features

3.1 Three-Tier Architecture

MedBooking follows a standard three-tier web architecture consisting of a React.js presentation layer, a Node.js application layer, and a MongoDB data layer.

3.2 Actor Model

Table 1 lists the primary actors based on stakeholder roles identified in the literature.

Table 1: System actors and literature basis

Actor	Role	Ref.
Patient	Books appointments; uses chatbot, telemedicine, health records	[8,3,4]
Doctor	Sets schedule; manages consultations; uses dashboard	[4]
Admin	Approves accounts; manages specialties; views statistics	[4]
Email System	External actor; executes automated notification commands	[3]

3.3 Core Functional Requirements

Table 2 presents the primary functional requirements, each grounded in the literature review.

Table 2: Core functional requirements with research basis

ID	Feature	Description	Ref.
FR-01	Registration	Email + password sign-up with OTP verification	—
FR-05	Relative mgmt.	Proxy booking for family members (elderly support)	[5]
FR-10	Doctor search	Filter by specialty, name, hospital, GPS location	[8]
FR-11	Slot display	Real-time slot visibility with controlled subset	[2]
FR-16	Auto reminder	Email + SMS notifications; one-click cancel/reschedule	[3]
FR-30	Video call	WebRTC real-time video consultation	[1,6]
FR-33	AI Chatbot	Symptom triage → specialty recommendation	[6,7]

3.4 Competitive Analysis

Table 3 so sánh tính năng của hệ thống với các đối thủ cạnh tranh tại Việt Nam.

4 Discussion

4.1 Limitations

Several limitations should be acknowledged. First, the system has not yet undergone comprehensive user testing with real patients and doctors; adoption metrics remain projected from the literature [1,4]. Second, the AI chatbot’s specialty classification accuracy target has not been empirically validated in the Vietnamese medical context. Third, load testing for high concurrent usage is

Table 3: Feature comparison with competing Vietnamese platforms

Platform	Booking	Telemd.	AI Bot	E- Pharm.	Map	OCR
BookingCare	✓	—	—	—	✓	—
eDoctor	✓	✓	—	✓	—	—
MedBooking	✓	✓	✓	✓	✓	✓

planned but not yet executed. Finally, the OCR prescription pipeline relies on Tesseract.js, which may underperform on handwritten notes.

5 Conclusion

This paper presented MedBooking, a patient-centred online medical appointment booking system developed for the SWP391 capstone course. The design is closely informed by a structured review of peer-reviewed papers spanning telemedicine adoption, scheduling optimisation, reminder systems, and AI implementation in healthcare. Future work will focus on usability testing with real clinics and benchmarking the AI classification algorithms against active clinical datasets.

References

1. He, H., Raja Ghazilla, R.A., Abdul-Rashid, S.H.: Factors influencing the intention to use telemedicine services among older adults in China. *Scientific Reports* **15**(1), 37772 (2025), <https://www.nature.com/articles/s41598-025-14630-8>
2. Kaluza, I., Voigt, G., Haase, K., Dietze, A.: Control of Online-Appointment Systems When the Booking Status Signals Quality of Service. *Schmalenbach Journal of Business Research* **76**(3), 397–432 (Sep 2024). <https://doi.org/10.1007/s41471-024-00188-0>, <https://link.springer.com/10.1007/s41471-024-00188-0>
3. Mohammed Selim, S., Senanayake, S., McPhail, S.M., Carter, H.E., Naicker, S., Kularatna, S.: Consumer Preferences for a Healthcare Appointment Reminder in Australia: A Discrete Choice Experiment. *The Patient - Patient-Centered Outcomes Research* **17**(5), 537–550 (Sep 2024). <https://doi.org/10.1007/s40271-024-00692-9>, <https://link.springer.com/10.1007/s40271-024-00692-9>
4. Ostadmohammadi, F., Nabovati, E., Jeddi, F.R., Arani, L.S.: Stakeholders’ experiences, perceptions and satisfaction with an electronic appointment system: a qualitative content analysis. *BMC Health Services Research* **25**(1), 220 (Feb 2025). <https://doi.org/10.1186/s12913-025-12289-5>, <https://bmchealthservres.biomedcentral.com/articles/10.1186/s12913-025-12289-5>
5. Rafi, S.: Artificial intelligence and telemedicine in elderly healthcare: A mixed-methods study. *BMC Geriatrics* **26**(1), 71 (Dec 2025). <https://doi.org/10.1186/s12877-025-06872-y>, <https://link.springer.com/10.1186/s12877-025-06872-y>

6. Sharma, P.: Smart healthcare: the role of AI, robotics, and NLP in advancing telemedicine and remote patient monitoring. *BMC Artificial Intelligence* **1**(1), 14 (Nov 2025). <https://doi.org/10.1186/s44398-025-00015-4>, <https://bmcartificialintel.biomedcentral.com/articles/10.1186/s44398-025-00015-4>
7. Silva-Ferreira, M., Cruz, S., Luís, M.S., Silva, M.C., Monteiro-Reis, S., Henrique, R., Jerónimo, C., Lefèvre, S.C., Laplaud, A., Frasca, M., Pollet, L., Zurbanobeaskoetxea, L., Barbastro, R.A., García, M.I.H., Galán, B.J., Palau, F.G., Marques, D.F., Durán, R.L.: Mapping study on AI-based technologies in palliative care – a scoping study. *BMC Palliative Care* **24**(1), 274 (Oct 2025). <https://doi.org/10.1186/s12904-025-01909-w>, <https://bmcpalliatcare.biomedcentral.com/articles/10.1186/s12904-025-01909-w>
8. Singer, C., Boldor, N., Vaknin, S., Olmer, L., Wilf-Miron, R., Myers, V.: Scheduling an appointment for MRI: patient perception of wait time and difficulty. *Israel Journal of Health Policy Research* **14**(1), 14 (Mar 2025). <https://doi.org/10.1186/s13584-025-00677-5>, <https://ijhpr.biomedcentral.com/articles/10.1186/s13584-025-00677-5>